

Madhuri Pyreddy

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EDUCATION

California State University Fresno

Fresno, California

B.S. Computer Science

08/2017 - 05/2021

- Courses: Algorithms and Data Structures, Big Data Analytics, Database Systems, Web Programming, Software Engineering, Simulation, Artificial Intelligence, Design and Analysis of Algorithms

Clovis North High School

Fresno, California

High School Diploma

08/2013 - 06/2017

TECHNICAL SKILLS

- Languages: Python, Angular, Java, C++
- Tools: Little bit of operational experience in Linux, Docker, Red Hat. Proficient with GitHub and Gitlab.

PROFESSIONAL EXPERIENCE

Alutiiq: Naval Base Ventura County Port Hueneme

Port Hueneme, California

Software Engineer, UOC-CCS (CSCAPE)

08/2024-03/2025

- Design software systems for Unmanned Operations to meet security, performance, scalability, reliability, and maintainability requirements. Primary responsibility is to Design for Test/Design for Debug (DFT/DFD).
- Required to communicate with the software development teams to troubleshoot applications by providing appropriate level of technical details and required to communicate with Fleet Operators to capture relevant data for the software development team.
- Document technical solutions and articulate these solutions to both team members and customers.
- Provide support to update and field applications real-time on both ashore and afloat data systems.
- Travel to remote sites/ships as required to meet operational requirements.
- Analyze current technology environment to detect critical deficiencies and recommend solutions for improvement.

Naval Undersea Warfare Center Keyport

Keyport, Washington

Computer Scientist, UOC-CCS (CSCAPE), RDMS

08/2022 - 02/2024

- Implemented front end code in Angular that enabled data types (METAR, Buoy, and Ship) to appear/disappear in new drawn data area on the geo, hovering information over and inspecting history on number of hours inputted for the data types on the geo; tested different data types in the data Area in the geo on Docker container and writing a report of the results
- Authored technical "how to" papers on enabling CAC readers in Firefox and Google on RHEL 8 Virtual Machine and installation of Google in RHEL 8 Virtual Machine and proposed UI Designs for METAR, Buoy, and Ship Hovers to PSE, and information of data types in the project itself to PSE (Pacific Science and Engineering)
- Scripted code that reads RSDF file and sends RSDF data from data file between client and server using TCP Connection in virtual machines (VMs) in Java and Python; implemented the same code as above in the ReplayServer Swing GUI in Java

Naval Undersea Warfare Center Keyport

Keyport, Washington

TRMC Stem Intern, Buoy Stations and Ships Observations

06/2021 - 08/2021

- Generated maps that plot the information stations and ships in Tkinter desktop application by reading the data file from NDBC (National Data Buoy Center) and using different Python libraries (tkinter, folium, webbrowser, pandas, markercluster)

PROJECTS

- PDMS (Patient Doctor Management System): Coordinated with a team of 3 to design software that help patients not only schedule their appointments with doctors online, but get recommended dosages to get better in another Android Studio (Java) mobile based application
- MyCar App: Cooperated with a team of 4 to design software that allows a customer to securely log in and rent a vehicle directly from a dealership based on personal preferences for the purpose of entertainment and/or transportation in Android Studio (Java) mobile based software application
- The Study of Weather Effects on COVID-19: Collaborated with a team of 6 to generate Python scripts that analyzes weather effects with mortality rate to see if weather can have an effect on COVID-19 given data-sets COVID-19 mortality rate, and weather aspects such as temperature, ozone levels, and humidity when the virus was present.

Misc.

- GMIS Stem Conference - Honorable Mention for Video Contest (promoting computer science education)
- Valley Children's Pediatric Hackathon - Awarded as Most Emerging Team for Hackathon